Yale University
UCSF
AI in Biology Symposium

# Deduce the World: How Far Can Reasoning Go?

Jeffrey Wei  Brian Cheung

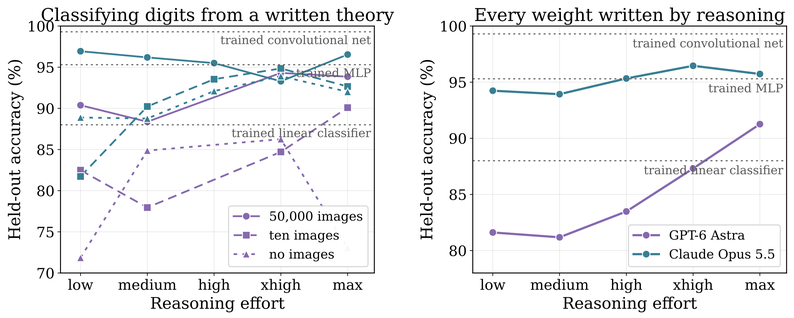
Results & figures

**TLDR:** We replace the learner of deep learning with a language model that must state an explicit theory, write it as code, and revise it against validation. The theories it forms predict competitively with tuned deep learning models, and generalize better to conditions held out from training.

## 1 Theories instead of fitted predictors

Scientific progress comes from proposing a theory, testing its predictions, and revising it where they fail. Deep learning does not work this way. It fits a predictor to existing data and cannot extrapolate beyond it.

| | |
|---|---|
| **May** | read papers and the web, measure the training data, inspect its own errors, revise its account |
| **May not** | fit parameters to labels, search automatically over rules, store labelled examples, copy weights, or see the test split |
| **Scored** | once, frozen, on held-out data |

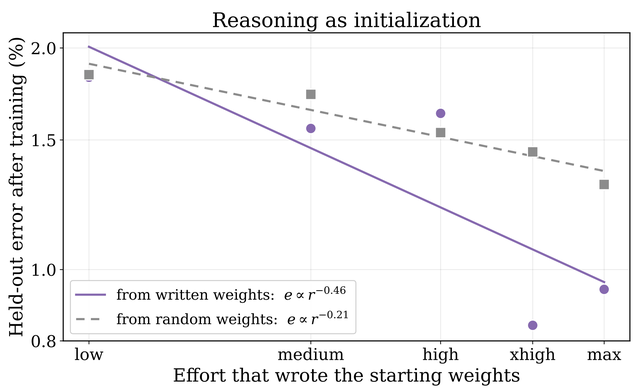



Handwritten digits. Left: accuracy given the full training set, one image per digit, or none at all. Right: networks whose every weight was written by reasoning, with no optimizer. Dotted lines are trained models.

Given no training images at all, a theory of how digits are written classifies most of them.

### Reasoning as initialization

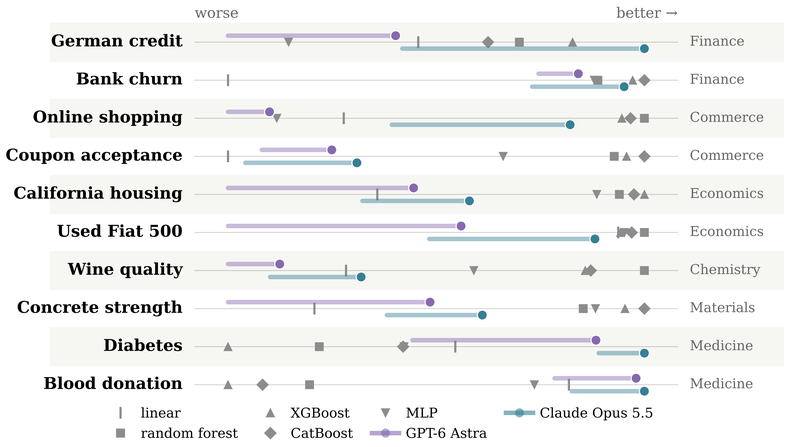
Those written weights are also a starting point. Training from them converges sooner, by more as reasoning grows.




Held-out error after a fixed training budget against effort rank, both axes logarithmic.

## 2 The same protocol, ten fields

Ten tabular tasks with fixed splits, against linear models, random forests, XGBoost, CatBoost and an MLP tuned on the same rows.




Each task on its own scale, oriented so that right is better. Bars span five reasoning efforts and the dot is the best of them. Grey marks are the tuned learned models.

Reasoned theories win on small clinical tables and trail boosted trees where the signal lies in feature interactions. The boundary belongs to the domain, not the model.

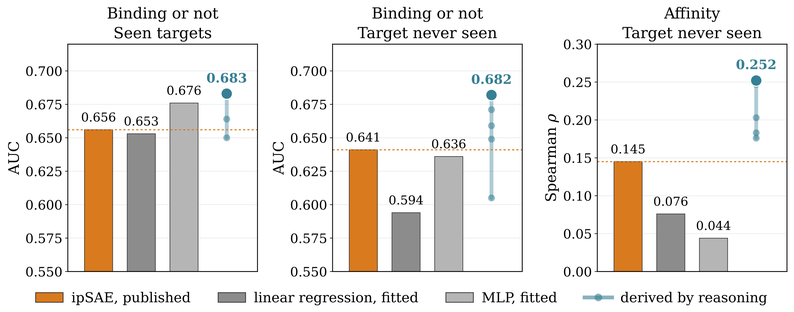
### What a theory looks like

Selected rules from one program on German credit. The points are judgments stated before measurement, not fitted odds ratios.

| | |
|---|---|
| Checking account | none $+3$, at least 200 DM $+1$, negative $-2$ |
| Credit history | critical account elsewhere $+2$, past delay $-1$ |
| Loan duration | $-1$ for each twelve months beyond the first twelve |
| Employment | under a year $-1$, four years or more $+1$ |
| Age | $-1$ below twenty-five |
| Output | $P(\text{good}) = 1/(1+e^{-(\text{score}-1)/2})$ |

## 3 Deriving a scientific score

A structure predictor returns a complex and an estimate of its own error. Each session receives that matrix and nothing else, exactly what the published score receives.



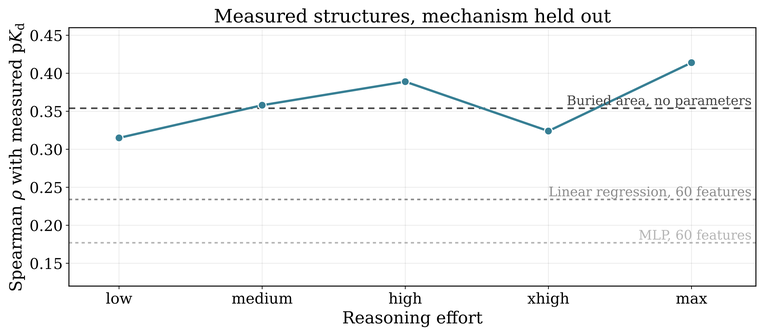

704 designed binders against five targets. The middle and right panels hold out an entire target.

### Derived, not fitted

Treating the bound complex as a harmonic well, with $\sigma_{\text{iface}}$ the precision of the interface and $\sigma_0$ the precision the binder achieves within itself,

$$\kappa = \left(\frac{\sigma_0}{\sigma_{\text{iface}}}\right)^2, \qquad \mathrm{p}K_{\mathrm{d}} = \mathrm{p}K_{\mathrm{d,ref}} + 3\log_{10}\kappa.$$

The 3 is half the six rigid-body degrees of freedom that association confines. Nothing is fitted.




Measured structures grouped by mechanism of association. One mechanism is held out whole and every mechanism takes a turn.

What transfers is a dimensionless ratio the pair supplies itself. What fails is anything calibrated on the cases it was fitted to.

Caveats: these models have read the literature of every domain we test, so recall and derivation are not fully separable. The protein cohort is curated candidates rather than a random draw, held-out counts are small and nothing is bootstrapped. The initialization result is one model family at a small seed count. An effort setting is not a measure of compute. Frozen protocols, programs and research logs are kept for every run and available on request.